

Rithwak Somepalli

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EDUCATION

Georgia State University

B.S. in Computer Science, GPA: 3.59

Atlanta, Georgia

Expected Graduation, May 2027

EXPERIENCE

Twin Brain, Evu

Remote

Software Development Engineering Intern

Jun 2025 – Aug 2025

- Integrated AI trading models with broker APIs (Alpaca, Interactive Brokers) to run large-scale simulations, handling 50,000+ requests/day during peak market-open traffic with zero downtime.
- Reduced incident response time by 60% through real-time health telemetry dashboards and automated alerts for abnormal trade patterns and model prediction drift.
- Partnered with other teams to get outputs of 99.9% data accuracy in a simulated environment.

PROJECTS

Heart Anomaly Detection

Dec 2025

- Designed a novel decreasing-width early-exit architecture in PyTorch, reducing energy usage by 35% while maintaining 95%+ accuracy by prioritizing logical reasoning over feature enumeration.
- Orchestrated a Kubernetes and Docker-based deployment on OCI, utilizing FastAPI and WebSockets to stream live EKG graphs with sub-200ms end-to-end latency via a CI/CD pipeline.

Cloud-Native Infrastructure & DevOps Pipeline

Nov 2025

- Automated full OCI cloud provisioning of 15+ resources with Terraform, implementing Python and Bash scripts to expose REST endpoints for health monitoring and status updates.
- Strengthened cloud security by implementing keyless authentication using OCI Instance Principals and Dynamic Groups, eliminating 100% of static credentials and passing a simulated security audit.

Co-Evolutionary Market Research (DRL vs EGT)

Aug - Oct 2025

- Architected a bid-ask market in Python using TensorFlow and Keras where a DRL agent and EGT populations competed over 100,000 episodes, with a 15% higher Sharpe ratio for DRL strategies.
- Utilized NumPy to validate results against 5 years of real market data, proving DRL systematically exploited evolutionary population stability thresholds.

Network Intrusion Detection System

Sept 2025

- Engineered an OCI-hosted NIDS using Node.js and Linux netfilter/iptables to perform Deep Packet Inspection on 10,000+ packets/minute, identifying DDoS, SQL injection, and port scanning with a <2% false positive rate.
- Mechanized real-time threat detection and analysis by streaming telemetry to a cloud-hosted REST endpoint, reducing the threat mitigation window to sub-second intervals.

SKILLS

Programming: Python, JavaScript, Java, C++, Bash, MySQL

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Node.js, Flask, Scikit-learn

DevOps & Cloud: Kubernetes, Docker, Terraform, Oracle Cloud, GitHub Actions, Linux (Systemd, Iptables)

Concepts: Microservices, REST APIs, CI/CD, Reinforcement Learning, Network Security, Zero-Trust Architecture

CERTIFICATIONS

Cybersecurity Certification

Google, May 2023